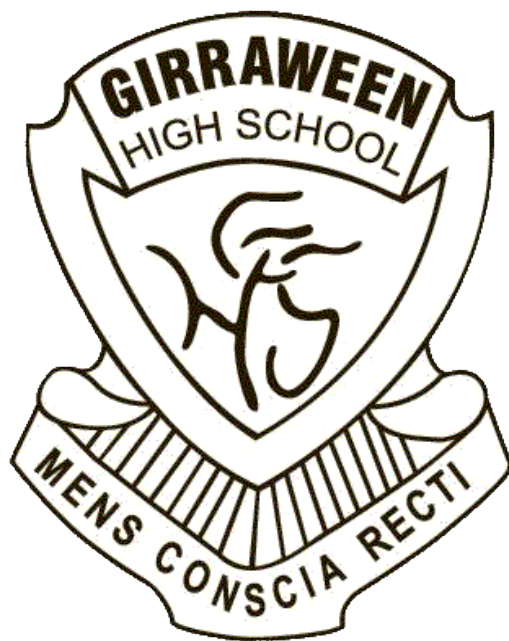




Girraween High School

2016 Year 12 Trial Higher School Certificate Mathematics Extension 1

General Instructions

- Reading Time - 5 minutes
- Working Time - 2 hours
- Calculators and ruler may be used
- All necessary working out must be shown
- Write on both sides of the paper

Total Marks - 70

- Attempt all questions
- Marks may be deducted for careless or badly arranged work

Section I

10 marks

Attempt Questions 1-10

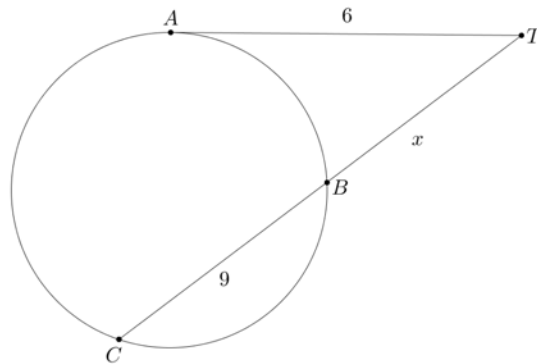
Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Question 1-10

Question 1 (1 mark)

Line AT is a tangent to the circle at A . TB is a secant cutting the circle at B and C .

If $AT = 6$, $TB = x$ and $BC = 9$, what is the value of x ?



- A. 2
- B. 3
- C. 4
- D. 12

Question 2 (1 mark)

What is the derivative of $\cos^{-1}(3x)$?

- A. $\frac{1}{3\sqrt{1-9x^2}}$
- B. $\frac{-1}{3\sqrt{1-9x^2}}$
- C. $\frac{3}{\sqrt{1-9x^2}}$
- D. $\frac{-3}{\sqrt{1-9x^2}}$

Question 3 (1 mark)

What is the value of $\lim_{x \rightarrow 0} \frac{2 \sin 3x}{x}$?

- A. $\frac{2}{3}$
- B. 2
- C. 6
- D. undefined

Question 4 (1 mark)

The point P divides the interval AB in the ratio $-1 : 2$. Which of the following diagrams is correct?



Question 5 (1 mark)

The degrees of two polynomials $P(x)$ and $Q(x)$ are m and n respectively, where $m > n$.

What is the degree of $P(x) + Q(x)$?

A. $m + n$

B. mn

C. m

D. n

Question 6 (1 mark)

What is the term independent of x in the expansion of $\left(x^2 - \frac{2}{x}\right)^9$?

A. ${}^9C_3(-2)^3$

B. ${}^9C_3(2)^3$

C. ${}^9C_6(-2)^6$

D. ${}^9C_6(2)^6$

Question 7 (1 mark)

A hotel has 3 different rooms.

How many different ways can 4 people be accommodated?

A. 3^4

B. 4^3

C. 4C_3

D. 4P_3

Question 8 (1 mark)

The position function of a particle is given by $x = 2 \sin \pi t + 1$. Which of the following is true?

- A. The maximum velocity of the particle occurs at $t = \frac{1}{2}$ at $x = 3$
- B. The maximum velocity of the particle occurs at $t = \frac{3}{2}$ at $x = -1$
- C. The maximum velocity of the particle occurs at $t = 2$ at $x = 1$
- D. The maximum velocity of the particle occurs at $t = 6$ at $x = -1$

Question 9 (1 mark)

For which of the following is true?

- A. If $f(x) = \sin x$ for $0 \leq x \leq \pi$ then $f^{-1}(x)$ exists
- B. If $f(x) = x^2$ for all real x then $f^{-1}(x)$ exists
- C. If $f(x) = mx$ for all real x then $f^{-1}(x)$ exists for any real value of m
- D. $f^{-1}(x)$ does not exist for any of the above

Question 10 (1 mark)

Projectiles A and B are launched at same time at velocity V and angle α . However projectile A is launched from a higher position. The two projectiles land in the same horizontal plane. Which of the following is always true?

- A. A and B will reach the ground at the same time
- B. A and B will have the same range
- C. A will reach its maximum height earlier than B
- D. The maximum speed of A is greater than the maximum speed of B

Question 11 on the next page

Section II

60 marks

Attempt Questions 11-14

Allow about 1 hour and 45 minutes for this section

Write your answers on the paper provided.

In Questions 11-14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks)

- (a) Solve $\frac{3}{x+2} < 4$. [3]
- (b) Find the size of the acute angle between the lines $2x + y = 5$ and $3x - y = 1$. [3]
- (c) The point P divides the interval joining $A(-1, -2)$ to $B(9, 3)$ internally in the ratio $4 : 1$. Find the coordinates of P . [2]
- (d) If $\cos \theta = \frac{4}{5}$ and $\frac{3\pi}{2} < \theta < 2\pi$ find the exact value of $\sin 2\theta$. [2]
- (e) Using the substitution $u = \sqrt{x}$ find $\int \frac{1}{\sqrt{x}(1+x)} dx$. [3]
- (f) Find the value of a if $P(x) = x^3 + ax^2 + ax + 5$ gives the same remainders when it is divided by $x + 2$ or $x - 4$. [2]

The exam continues on the next page

Question 12 (15 marks)

- (a) Prove by mathematical induction that for all integers $n \geq 1$, [3]

$$1 \times 2^0 + 2 \times 2^1 + 3 \times 2^2 + \cdots + n \times 2^{n-1} = 1 + (n-1)2^n$$

- (b) i. Find the coefficient of x in the expansion of $\left(3x - \frac{1}{2x}\right)^7$. [2]

- ii. Hence state the constant term in the expansion of $\left(1 + \frac{1}{x}\right)^2 \left(3x - \frac{1}{2x}\right)^7$. [1]

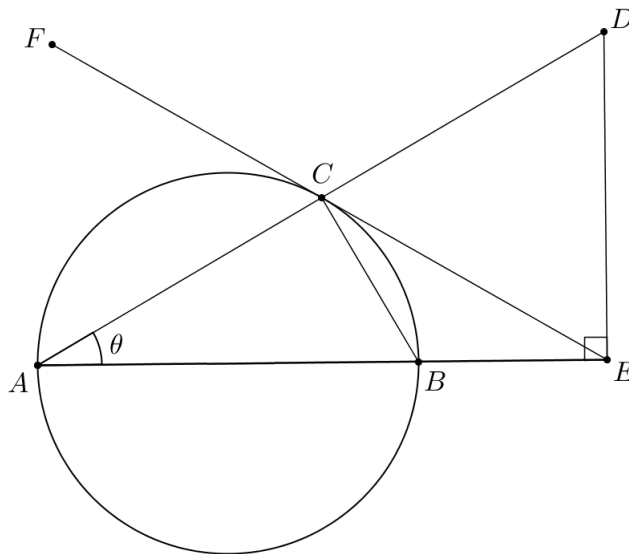
- (c) Consider the function $f(x) = x^3 + x$.

- i. State the domain of $f(x)$. [1]

- ii. Show that $f(x)$ does not have any stationary points. [2]

- iii. By giving reasons, state whether $f^{-1}(x)$ exists. [1]

- (d) The diagram shows a circle with diameter AB . EF is a tangent to the circle at C . AC is extended to D such that DE is perpendicular to AE . Let $\angle BAC = \theta$.



- i. Copy the diagram and prove that $BCDE$ is a cyclic quadrilateral. [2]

- ii. Prove that $EC = ED$. [3]

The exam continues on the next page

Question 13 (15 marks)

- (a) i. A fair coin is tossed 4 times, what is probability of getting more heads than tails? [2]

- ii. A person decides to flip a two dollar coin 4 times each day, and if he gets more heads than tails he will contribute the coin towards his savings. He does this for 7 days. What is the probability that he will contribute four dollars towards his savings by the end of the 7 days? Give your answer to one decimal place. [1]

- (b) A particle moves along a straight line with displacement x m and velocity v ms^{-1} . Initially the particle is at the origin at with velocity -1 ms^{-1} . The acceleration of a particle is given by

$$\ddot{x} = 4x + 2$$

- i. Show that $v^2 = 4x^2 + 4x + 1$ [2]

- ii. Show that $x = \frac{1}{2}(e^{-2t} - 1)$ [3]

- iii. What happens to x as $t \rightarrow \infty$? [1]

- (c) A particle moves in a straight line and its position at time t is given by

$$x = 5 + \sqrt{3} \sin 3t - \cos 3t$$

- i. Express $\sqrt{3} \sin 3t - \cos 3t$ in the form $R \sin (3t - \alpha)$, where α is in radians. [2]

- ii. Prove that the particle is undergoing simple harmonic motion and find its period. [2]

- iii. State the particle's maximum displacement. [1]

- iv. When does the particle first reach its minimum acceleration? [1]

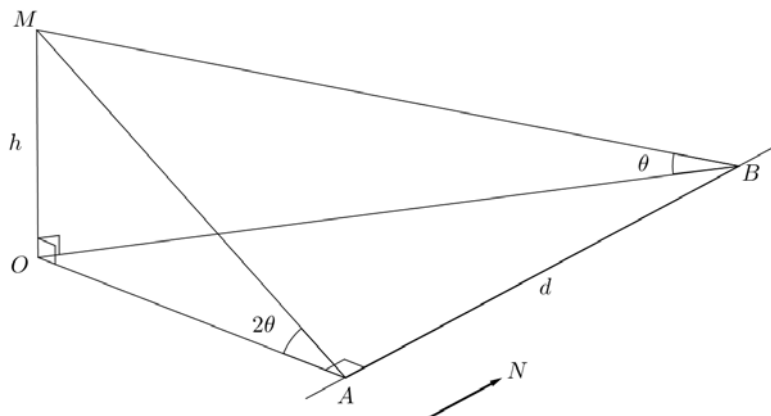
The exam continues on the next page

Question 14 (15 marks)

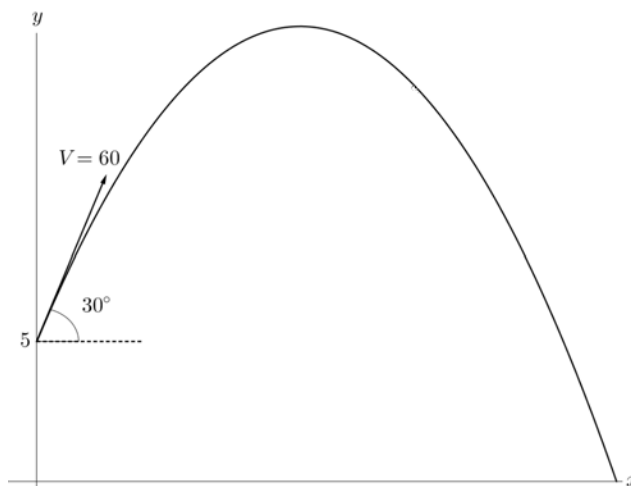
- (a) A person walks a length of d metres due north along a road from point A to point B . The point A is due east of a mountain OM , where M is the top of the mountain. The point O is directly below point M and is on the same horizontal plane as the road. The height of the mountain above point O is h metres.

From point A , the angle of elevation to the top of the mountain is 2θ .

From point B , the angle of elevation to the top of the mountain is θ .



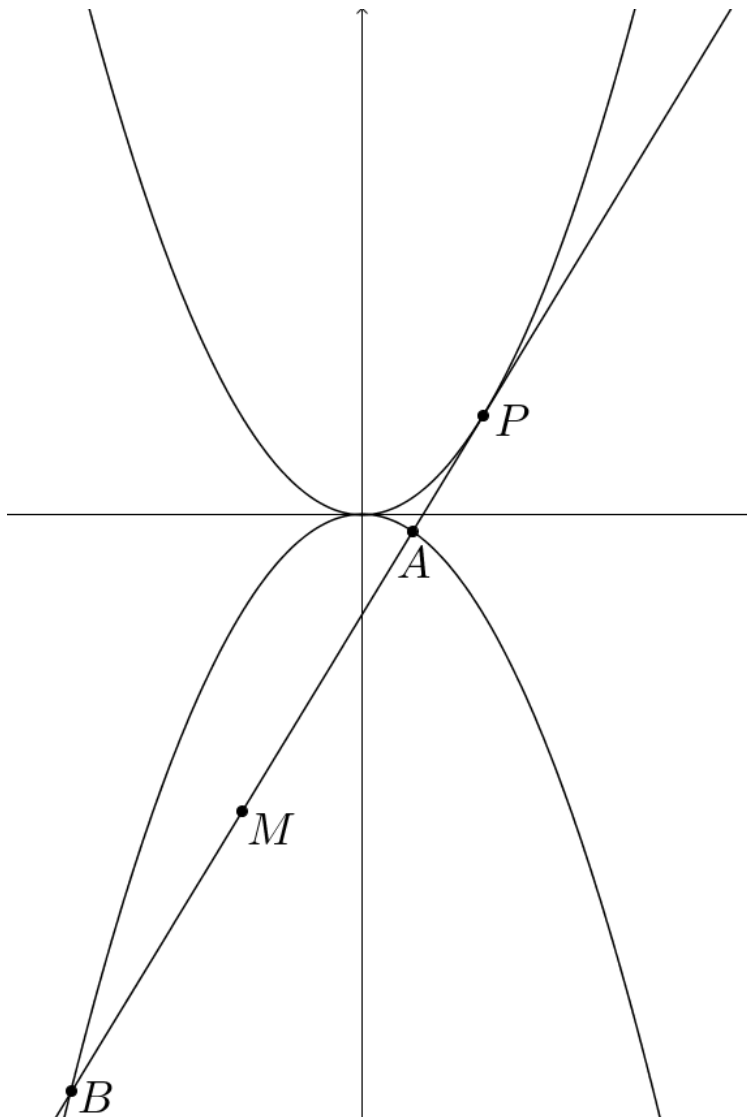
- i. Find the expressions for OA and OB in terms of h and θ . [1]
 - ii. Show that $d^2 = \frac{h^2 \operatorname{cosec}^2 \theta}{4} (3 - \tan^2 \theta)$ [3]
- (b) A projectile is launched from a height of 5 m above the ground at $V = 60\text{ ms}^{-1}$ at 30° to the horizontal. You may assume $g = 10\text{ ms}^{-2}$.



- i. Derive the equations for x and y . [2]
- ii. Find the maximum height of the projectile. [1]
- iii. Find the speed and angle (to nearest degree) at which the projectile hits the ground. [3]

The exam continues on the next page

- (c) The diagram shows a point $P(2at, at^2)$ on the parabola $x^2 = 4ay$. The tangent to the parabola at P cuts the reflection of the parabola in the x axis at points A and B . The point M is the midpoint of the interval AB .



- i. Show that the equation of the tangent is $y = tx - at^2$. [1]
- ii. Show that the coordinates of M are $(-2at, -3at^2)$ [3]
- iii. Show that the locus of M is $x^2 = -\frac{4}{3}ay$ [1]

End of exam

Y12 Ext1 TRIAL 2016

MC: BDCAC DACDD
C

Q1

$$6^2 = n(n+9)$$

$$36 = n^2 + 9n$$

$$n^2 + 9n - 36 = 0$$

$$(n+12)(n-3) = 0$$

$$\therefore n = 3 \quad \therefore \textcircled{B}$$

Q2 $y = \cos^{-1}(3x)$

$$y' = -\frac{1}{\sqrt{1-(3x)^2}} \times 3$$

$$= -\frac{3}{\sqrt{1-9x^2}} \quad \therefore \textcircled{D}$$

Q3 $\lim_{x \rightarrow 0} \frac{25 \sin 3x}{x} = 2 \lim_{x \rightarrow 0} \frac{\sin 3x}{x}$

$$= 2 \times 3 \lim_{x \rightarrow 0} \frac{\sin 3x}{3x}$$
$$= 6 \lim_{x \rightarrow 0} \frac{\sin 3x}{3x}$$
$$= 6 \quad \therefore \textcircled{E}$$

Q4

$$AP: PB = -1: 2$$

$\therefore P$ is external to AB
and P is closer to A

$$\therefore \textcircled{A}$$

Q5

$$p(n) = a_m n^m + a_{m-1} n^{m-1} + \dots + a_0$$

$$q(n) = b_n n^n + b_{n-1} n^{n-1} + \dots + b_0$$

$$\therefore \deg[p(n) + q(n)] = m \quad \therefore \textcircled{C}$$

Q6

Constant term occurs when

$$\left(\frac{9}{3}\right) (x^2)^3 \left(-\frac{2}{x}\right)^6$$

$$= \left(\frac{9}{3}\right) (-2)^6 \frac{x^6}{x^6}$$

$$= \left(\frac{9}{3}\right) 2^6 \quad \therefore \textcircled{D} \text{ or } \textcircled{C}$$

Q7

Each person can choose one
of the 3 rooms $\therefore 3 \times 3 \times 3 \times 3$
 $= 3^4 \quad \therefore \textcircled{A}$

Q8



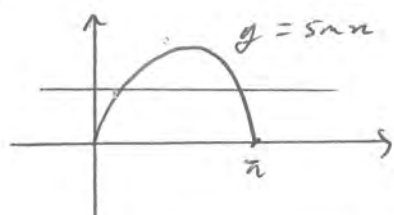
$\therefore$ max V occurs at $x=1$

$$T = \frac{2x}{x} = 2$$

$\therefore$ max V occurs when $t=0$
as it is at the centre initially

$\therefore$ it will return to this
position at $t=2 \quad \therefore \textcircled{C}$

Q9



clearly not 1:1

$\therefore$ A is false.

$f(x) = x^2$ is clearly even as

$$f(-x) = (-x)^2 = x^2 = f(x).$$

As even functions are symmetrical about the y-axis they are not 1:1

$\therefore$ B is false

If $m=0$ then $f(x) = mx$ is a horizontal line which is clearly not 1:1 $\therefore$ C is false

$\therefore$ (D)

Q10

Since $y = V \sin \alpha - gt$ for both projectiles they will reach the highest points at the same time

$\therefore$ C is false

A clearly has a longer flight time as it needs to fall a greater vertical distance $\therefore$ A is false
With longer flight time A

must travel a longer horizontal distance $\therefore$ B is false.

$\therefore$ (D) is true as it will gain a more negative vertical velocity, resulting in greater maximum speed.

Q11

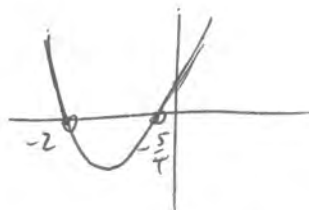
(a) $\frac{3}{x+2} < 4$

$$3(x+2) < 4(x+2)^2$$

$$4(x+2)^2 - 3(x+2) > 0$$

$$(x+2)[4(x+2) - 3] > 0$$

$$(x+2)(4x+5) > 0$$



$$\therefore x < -2 \text{ \& } x > -\frac{5}{4}.$$

(b) $y = -2x + 5 \therefore m_1 = -2$

$y = 3x + 1 \therefore m_2 = 3$

$$\tan \theta = \left| \frac{-2-3}{1-6} \right| = 1$$

$$\therefore \theta = 45^\circ$$

Q11

(c)

$$A(-1, -2) \quad B(9, 3)$$

$$\begin{array}{c} \times \\ 4 : 1 \end{array}$$

$$x = \frac{4 \times 9 - 1}{4 + 1} \quad y = \frac{3 \times 4 - 2}{5}$$

$$x = 7 \quad y = 2$$

$$\therefore P = (7, 2)$$

(d)

$$\cos \theta = \frac{4}{5}$$



$$\text{Since } \frac{3\pi}{2} < \theta < 2\pi \therefore \sin \theta = -\frac{3}{5}$$

$$\therefore \sin 2\theta = 2 \sin \theta \cos \theta$$

$$= 2 \times -\frac{3}{5} \times \frac{4}{5} = -\frac{24}{25}$$

(e)

$$I = \int \frac{1}{\sqrt{n}(1+n)} dn$$

$$\text{Let } u = \sqrt{n}$$

$$u = n^{\frac{1}{2}}$$

$$du = \frac{1}{2} n^{-\frac{1}{2}} dn$$

$$du = \frac{1}{2\sqrt{n}} dn$$

$$\therefore I = 2 \int \frac{1}{2\sqrt{n}(1+n)} dn$$

$$= 2 \int \frac{1}{1+u^2} du$$

$$= 2 \tan^{-1} u + C$$

$$= 2 \tan^{-1} \sqrt{n} + C$$

(f)

$$p(-2) = p(4)$$

$$\therefore (-2)^3 + 4a - 2a + 5 = 4^3 + 16a + 4a + 5$$

$$-8 + 2a = 64 + 20a$$

$$\therefore 18a = -72$$

$$\therefore a = -4$$

Q12

(a)

• Prove true for $n=1$.

$$\text{When } n=1 : \text{LHS} = 1 \times 2^0 = 1$$

$$\text{RHS} = 1 + 0 \times 2^1 = 1$$

$$\therefore \text{LHS} = \text{RHS} = 1$$

$\therefore$ true for $n=1$

• Assume true for $n=k$, i.e.

$$1 \times 2^0 + 2 \times 2^1 + 3 \times 2^2 + \dots + k \times 2^{k-1}$$

$$= 1 + (k-1)2^k$$

• Prove true for $n=k+1$, i.e.

$$1 \times 2^0 + 2 \times 2^1 + \dots + k \times 2^{k-1} + (k+1) \times 2^k$$

$$= 1 + k \times 2^{k-1}$$

$$\text{LHS} = 1 \times 2^0 + 2 \times 2^1 + \dots + k \times 2^{k-1} + (k+1) \times 2^k$$

$$= 1 + (k-1)2^k + (k+1) \times 2^k \text{ by assumption.}$$

$$= 1 + 2^k (k-1 + k+1)$$

$$= 1 + 2^k (2k)$$

$$= 1 + k \times 2^{k+1} = \text{RHS.}$$

in - time for next

49

$$\begin{aligned} (i) \quad T_{n+1} &= \binom{7}{k} (3n)^{7-k} \left(-\frac{1}{2n}\right)^k \\ &= \binom{7}{k} 3^{7-k} n^{7-k} (-2)^k n^{-k} \\ &= \binom{7}{k} 3^{7-k} (-2)^k n^{7-2k} \end{aligned}$$

$$\therefore 7 - 2k = 1 \quad \therefore 2k = 6 \quad \therefore k = 3$$

$$\therefore \text{Coefficient is } \binom{7}{3} 3^4 (-2)^3$$
$$= -\frac{2835}{8}$$

$$(1 + \frac{2}{n} + \frac{1}{n^2}) (3n - \frac{1}{n})^7$$

only way to get constant is
multiplying $\frac{2}{n}$ with n term from
 $(3n - \frac{1}{2n})^7$

$\therefore \text{Constant } D \quad 2x - \frac{2835}{8} = -\frac{2835}{4}$

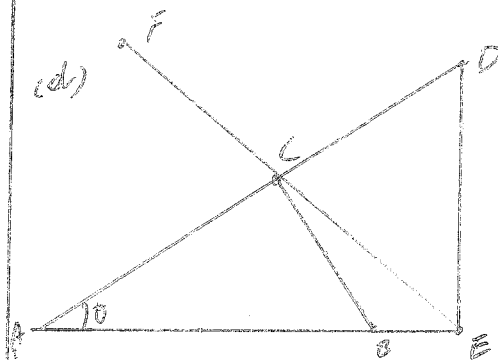
(c) (i) $\forall$ all real n .

(ii) $f'(n) = 3n^2 + 1 \neq 0$ for all n .

(iii) $f'(n) \geq 1$ for all n

$\therefore f$ is a monotonically increasing function, so it is 1-1

$\therefore f^{-1}$ exists.



(i) $\angle ACB = 90^\circ$ ($\angle$ in a semi circle with diameter AB)

$\therefore \angle CBE = 90^\circ$ (exterior $\angle$ of $\triangle ABC$).

$$\angle ADE = 90^\circ - \angle C \text{ (sum of } \triangle ADE)$$

$$\therefore \angle CBE + \angle ADE = 180^\circ$$

$\therefore BCDE$ is a cyclic quadrilateral
(opposite $\angle$'s are supplementary).

(ii) $\angle ABC = 90^\circ$ ($\angle$ sum of $\triangle ABC$).

2. $\angle ACF = 90^\circ$ ($\angle$ between
tangent EF and
chord AC equals
the $\angle$ in the
alternate segment).

$\therefore \angle DCE = 90^\circ$ (vertically opposite $\angle$'s)

$$\therefore \angle DCE = \angle ADE = 90^\circ - \theta$$

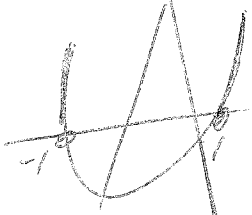
$\therefore CE = ED$ (Equal sides opposite equal $\angle$'s of $\triangle ECD$)

Q12

(a)

$$(1) \frac{n+1}{n-1} > 0$$

$$(n+1)(n-1) > 0$$



$$\therefore D: n < -1 \text{ \& } n > 1$$

or)

$$f(n) = \ln \sqrt{\frac{n+1}{n-1}}$$

$$f(n) = \frac{1}{2} \ln \left(\frac{n+1}{n-1} \right)$$

$$f(n) = \frac{1}{2} \left[\ln(n+1) - \ln(n-1) \right]$$

$$f'(n) = \frac{1}{2} \left[\frac{1}{n+1} - \frac{1}{n-1} \right]$$

$$= \frac{1}{2} \left[\frac{n-1 - (n+1)}{(n+1)(n-1)} \right]$$

$$f'(n) = \frac{1}{2} \times \frac{-2}{(n+1)(n-1)} = -\frac{1}{(n+1)(n-1)}$$

$\neq 0$ for all n

$\therefore$ no stationary points.

(iii) Since $f'(n) < 0$ for all n

$\therefore f$ is a monotonically decreasing function, so it is 1:1

$\therefore f^{-1}$ exists.

Q13

(a)

(1)

$$= P(3H) + P(4T)$$

$$= \binom{4}{3} \left(\frac{1}{2}\right)^4 + \binom{4}{4} \left(\frac{1}{2}\right)^4$$

$$= \frac{5}{16}$$

$$\text{or) } \binom{7}{2} \left(\frac{5}{16}\right)^2 \left(\frac{11}{16}\right)^5$$

$$= 31.5\% \text{ (1 dp)}$$

(b)

$$(i) \frac{d}{dn} \frac{1}{2} v^2 = 4n+2$$

$$\therefore \frac{1}{2} v^2 = \int (4n+2) dn$$

$$\frac{1}{2} v^2 = 2n^2 + 2n + C$$

$$\text{but } v = -1 \text{ when } n = 0$$

$$\therefore \frac{1}{2} = C$$

$$\therefore \frac{1}{2} v^2 = 2n^2 + 2n + \frac{1}{2}$$

$$\therefore v^2 = 4n^2 + 4n + 1$$

$$(ii) v^2 = (2n+1)^2$$

$$\therefore v = \pm (2n+1)$$

$$\text{but } v = -1 \text{ when } n = 0$$

$$\therefore v = -(2n+1)$$

$$\therefore \frac{dn}{dt} = -(2n+1)$$

$$\therefore \frac{1}{2n+1} dn = -dt$$

Q13

(b)

$$(ii) \int \frac{1}{2n+1} dx = - \int dt$$

$$\frac{1}{2} \int \frac{2}{2n+1} dx = -t + C$$

$$\frac{1}{2} \ln(2n+1) = -t + C$$

$$t=0 \quad n=0$$

$$\therefore \frac{1}{2} \ln 1 = C \quad \therefore C=0$$

$$\therefore \ln(2n+1) = -2t$$

$$\therefore 2n+1 = e^{-2t}$$

$$\therefore n = \frac{1}{2}(e^{-2t} - 1)$$

$$(iii) \text{ As } t \rightarrow \infty \quad n \rightarrow -\frac{1}{2}$$

(c)

(i)

$$\sqrt{3} \sin 3t - \cos 3t = R \sin(3t - \alpha)$$

$$\sqrt{3} \sin 3t - \cos 3t = R [\sin 3t \cos \alpha - \cos 3t \sin \alpha]$$

$$\sqrt{3} \sin 3t - \cos 3t = R \cos \alpha \sin 3t - R \sin \alpha \cos 3t$$

$$\therefore R \cos \alpha = \sqrt{3} \quad \& \quad R \sin \alpha = 1$$

$$\therefore R^2 \cos^2 \alpha + R^2 \sin^2 \alpha = 3 + 1$$

$$\therefore R^2 (\sin^2 \alpha + \cos^2 \alpha) = 4$$

$$\therefore R^2 = 4 \quad \therefore R = 2$$

$$\therefore \cos \alpha = \frac{\sqrt{3}}{2} \quad \& \quad \sin \alpha = \frac{1}{2}$$

$$\therefore \alpha = \frac{\pi}{6}$$

$$\therefore \sqrt{3} \sin 3t - \cos 3t = 2 \sin(3t - \frac{\pi}{6})$$

$$(ii) x = 5 + 2 \sin(3t - \frac{\pi}{6})$$

$$\dot{x} = 6 \cos(3t - \frac{\pi}{6})$$

$$\ddot{x} = -18 \sin(3t - \frac{\pi}{6})$$

$$\ddot{x} = -3^2 \times 2 \sin(3t - \frac{\pi}{6})$$

$$\ddot{x} = -3^2(x-5)$$

$$\therefore \text{SHM with } T = \frac{2\pi}{3}$$

$$(iii) x_{\max} = 5 + 2 = 7$$

$$(iv) \text{ minimum acceleration}$$

$$\text{when } x = 7$$

$$\therefore 7 = 5 + 2 \sin(3t - \frac{\pi}{6})$$

$$\therefore \sin(3t - \frac{\pi}{6}) = 1$$

$$\therefore 3t - \frac{\pi}{6} = \frac{\pi}{2}$$

$$\therefore t = \frac{1}{3} \times \left(\frac{\pi}{2} + \frac{\pi}{6} \right)$$

$$t = \frac{1}{3} \times \frac{4\pi}{6} = \frac{2\pi}{9}$$

Q14

(a)

$$(i) \tan 2\theta = \frac{h}{OA}$$

$$\therefore OA = \frac{h}{\tan 2\theta}$$

$$\tan \theta = \frac{h}{OB}$$

$$\therefore OB = \frac{h}{\tan \theta}$$

Q14

(a)

$$(ii) d^2 = OB^2 - OA^2$$

$$= \frac{h^2}{\tan^2 \theta} - \frac{h^2}{\tan^2 2\theta}$$

$$= h^2 \left(\frac{1}{\tan^2 \theta} - \frac{1}{\tan^2 2\theta} \right)$$

$$= h^2 \left(\frac{1}{\tan^2 \theta} - \frac{(1 - \tan^2 \theta)^2}{4 \tan^2 \theta} \right)$$

$$= h^2 \left(\frac{4 - 1 + 2 \tan^2 \theta - \tan^4 \theta}{4 \tan^2 \theta} \right)$$

$$= h^2 \left(\frac{3 + 2 \tan^2 \theta - \tan^4 \theta}{4 \tan^2 \theta} \right)$$

$$= h^2 \frac{(3 - \tan^2 \theta)(1 + \tan^2 \theta)}{4 \tan^2 \theta}$$

$$= \frac{h^2 (3 - \tan^2 \theta)}{4 \tan^2 \theta \times \cos^2 \theta}$$

$$= \frac{h^2 \sec^2 \theta (3 - \tan^2 \theta)}{4}$$

(c) (14(b) on next page).

$$(i) g = \frac{1}{4a} n^2$$

$$g' = \frac{1}{2a} n$$

$$g'(2at) = t$$

$$\therefore g - at^2 = t(n - 2at)$$

$$g - at^2 = tn - 2at^2$$

$$g = tn - at^2$$

Q14

(c)

(ii) The reflection of the parabola is $x^2 = -4ay$.

So its intersection with the tangent is given by:

$$\frac{-x^2}{4a} = tn - at^2$$

$$-x^2 = 4atn - 4a^2 t^2$$

$$x^2 + 4atn - 4a^2 t^2 = 0$$

n coordinate of M is average of roots

$\therefore$ the n coordinate of M is

$$\text{given by } n = \frac{-4at}{2(1)} = -2at$$

$$\therefore g = t(-2at) - at^2$$

$$= -2at^2 - at^2 = -3at^2$$

$$\therefore M = (-2at, -3at^2)$$

$$(iii) \therefore t = -\frac{x}{2a}$$

$$\therefore g = -3a \left(-\frac{x}{2a} \right)^2$$

$$g = -3a \times \frac{x^2}{4a^2}$$

$$g = -\frac{3x^2}{4a}$$

$$\therefore x^2 = -\frac{4}{3}ay$$

$$x^2 = -4\left(\frac{a}{3}\right)y$$

$\therefore$ focal length = $\frac{1}{3}$ of original focal length.

Q14

(b) $\ddot{x} = 0$

(c) $\ddot{x} = \int 0 dt$

$$\dot{x} = C_1$$

but $t=0$ $\dot{x} = 60 \cos 30$

$$\dot{x} = 30\sqrt{3}$$

$$\therefore C_1 = 30\sqrt{3}$$

$$\therefore \dot{x} = 30\sqrt{3}$$

$$x = \int 30\sqrt{3} dt$$

$$x = 30\sqrt{3}t + C_2$$

but $t=0$ $x=0 \therefore C_2=0$

$$\therefore x = 30\sqrt{3}t$$

$$\ddot{y} = -10$$

$$\dot{y} = \int -10 dt$$

$$\therefore \dot{y} = -10t + C_3$$

$t=0$ $\dot{y} = 60 \sin 30$

$$\dot{y} = 30$$

$$\therefore C_3 = 30$$

$$\therefore \dot{y} = -10t + 30$$

$$y = \int -10t + 30 dt$$

$$y = -5t^2 + 30t + C_4$$

$t=0$ $y=5 \therefore C_4=5$

$$\therefore y = -5t^2 + 30t + 5$$

Q15

$$\dot{y}=0 \text{ when } -10t + 30 = 0$$

$$\therefore 10t = 30$$

$$t = 3$$

$$y(3) = -5(3)^2 + 30(3) + 5$$

$$= 50$$

$\therefore$ max height is 50 m.

(ii) $\dot{y}=0$ when:

$$-5t^2 + 30t + 5 = 0$$

$$t^2 - 6t - 1 = 0$$

$$t = \frac{6 \pm \sqrt{36 - 4(1)(-1)}}{2}$$

$$t = \frac{6 + \sqrt{40}}{2} = 3 + \sqrt{10}$$

$$\dot{y}(3 + \sqrt{10}) = -30 - 10\sqrt{10} + 30$$

$$= -10\sqrt{10}$$

$$v^2 = (30\sqrt{3})^2 + (-10\sqrt{10})^2$$

$$v^2 = 3700$$

$$\therefore v = 10\sqrt{37} \text{ ms}^{-1}$$

$$\tan \theta = \frac{10\sqrt{10}}{30\sqrt{3}} = \frac{\sqrt{10}}{3\sqrt{3}}$$

$$\therefore \theta = 31^\circ \text{ (below horz)}$$

$$\therefore v = 10\sqrt{37} \text{ ms}^{-1} \text{ at } 31^\circ \text{ below the horizontal.}$$